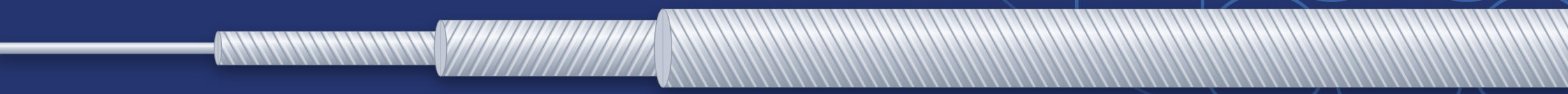
ALTAWAKOL
CABLES

Overhead Conductors

Product Catalogue

كتالوج الموصلات الهوائية



AAC | AAAC | ACSR

IEC 61089 · BS EN 50182 · ES

Table of Content

المحتويات

03	About Al Tawakol Group	مجموعة التوكل
04	Altawakol for Cables	التوكل للكابلات
05	Manufacturing Process	مراحل التصنيع
06	Quality & Testing	الجودة والاختبارات
07	Product Range	مجموعة المنتجات
08	AAC · All Aluminium	موصلات الألومنيوم
11	AAAC · Aluminium Alloy	موصلات سبائك الألومنيوم
14	ACSR · Steel Reinforced	موصلات الألومنيوم المقواة بالصلب
18	Technical Data	بيانات فنية
19	Packing & Drums	التعبئة والبكرات
20	Ordering Information	معلومات الطلب

مجموعة التوكل

WHO WE ARE

About Al Tawakol Group

Al Tawakol Group is a family-owned Egyptian business founded in 1975, built on the acquisition of Fratelli Gila — a trading house established in 1892 — and grown into a diversified industrial group.

Over five decades the group has expanded from electrical trading into manufacturing, contracting and investment across electrical solutions, steel fabrication and galvanization, renewable energy, trade and distribution. It operates more than ten factories in Egypt and works alongside world-leading partners including Schneider Electric, ACWA Power and Nexans.

Our values — **Excellence, Humanity, Accountability and Customer Focus** — have been carried across generations, and they guide every product that leaves our plants.

Mission

To always strive for success and commit to the value we create, by endeavoring to create better life solutions for everyone.

Vision

To always offer better life solutions, leaving a positive and genuine imprint on society.

45+
years in the Egyptian market

10+
factories in Egypt

180,000
sqm of factory area

2,400+
employees

300+
engineers

5+
ISO certifications



OUR FACTORY

Altawakol for Cables

التوكل للكابلات

Altawakol for Cables is the group's cable-manufacturing company, producing bare overhead conductors and low-voltage cables for Egypt's distribution utilities, contractors and EPC partners, and for export.

Our overhead-line range covers all-aluminium (AAC), all-aluminium-alloy (AAAC) and aluminium conductor steel-reinforced (ACSR) conductors, manufactured to IEC 61089, BS EN 50182 and the relevant Egyptian Standards — or to the customer's own specification.

Backed by the engineering depth of Al Tawakol Group — from power panels and substations to transmission towers and galvanization — we understand the network our conductors are built for.

What we offer

- AAC, AAAC and ACSR from 16 mm² to 1000 mm²
- Metric and BS code-name sizes in one range
- Manufacture to IEC, BS EN, ES or customer specification
- Type, sample and routine tests with full reports
- Wooden or steel drums, lengths to suit your stringing plan
- Support for conductor selection and sag–tension

HOW WE MAKE IT

Manufacturing Process

مراحل التصنيع

Every conductor follows a controlled route from rod to drum, with in-process checks at each stage and final testing before dispatch.

01

Aluminium rod

Electrical-grade aluminium (EC/1350) or Al-Mg-Si alloy rod is received, sampled and checked for chemistry and conductivity.

02

Wire drawing

Rod is drawn through multi-die machines to the exact wire diameter, with continuous diameter and surface control.

03

Heat treatment

Alloy wires for AAAC are precipitation-hardened (aged) to reach the strength and conductivity required by EN 50183 / IEC 60104.

04

Steel core

Galvanized steel wires for ACSR are inspected for tensile strength, elongation and zinc-coating mass before stranding.

05

Stranding

Wires are laid in concentric layers on rigid stranders with controlled lay ratios, alternating direction and a right-hand outer layer.

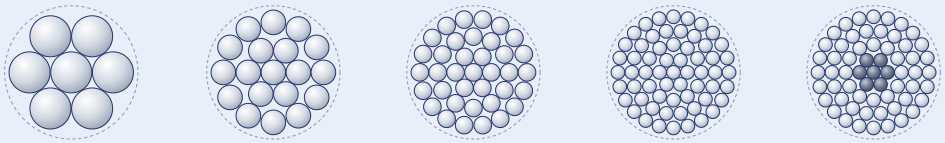
06

Testing & drumming

Finished conductor is measured, tested, wound under controlled tension onto drums, sealed, lagged and labelled.

Stranding geometry

concentric layers, 1+6+12+18+...



BUILT TO STANDARD

Quality & Testing

الجودة والاختبارات

Quality is designed into each stage of production. Raw materials, wires and finished conductors are tested in our laboratory against the requirements of the applicable standard, and every drum is shipped with its test certificate.

Applicable standards

IEC 61089	Round wire concentric lay overhead electrical stranded conductors
BS EN 50182	Conductors for overhead lines — round wire concentric lay stranded conductors
IEC 60889	Hard-drawn aluminium wire for overhead line conductors (AL1)
EN 50183 / IEC 60104	Aluminium-magnesium-silicon alloy wire (AL2 / AL3)
EN 50189 / IEC 60888	Zinc-coated steel wires for stranded conductors (ST1A)
ES	Egyptian Standards for bare overhead conductors
ASTM	B230/B231, B398/B399, B232 — on request

Wire tests

- Diameter and surface condition
- Tensile strength and elongation
- Resistivity / conductivity
- Wrapping and torsion (ductility)
- Zinc-coating mass and adherence (steel)

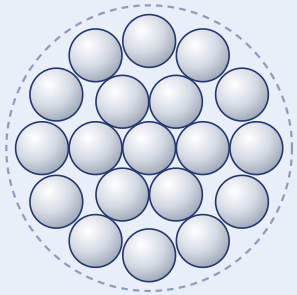
Conductor tests

- Lay ratio and lay direction
- Mass per unit length and overall diameter
- DC resistance at 20 °C
- Rated tensile strength (type test)
- Stress–strain characteristics (type test)

OVERVIEW

Product Range

مجموعة المنتجات



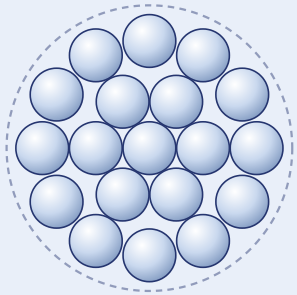
AAC

page 08

All Aluminium Conductor

Hard-drawn AL1 / 1350 aluminium wires. Highest conductivity per kilogram for short spans, urban distribution and substation busbars.

16 – 1000 mm²



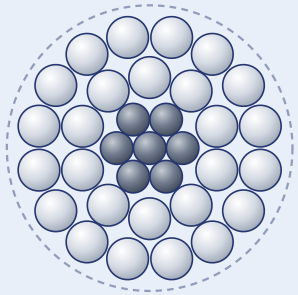
AAAC

page 11

All Aluminium Alloy Conductor

Al-Mg-Si alloy (AL3) wires. Higher strength and lower sag than AAC, with excellent corrosion resistance — ideal for coastal and desert lines.

16 – 1000 mm²



ACSR

page 14

Aluminium Conductor Steel Reinforced

AL1 wires stranded over a galvanized steel core. High tensile strength for long spans, river crossings and transmission lines.

16/2.5 – 490/65 mm²

Selection guide	AAC	AAAC	ACSR
Electrical conductivity	●●●●	●●●○	●●●○
Tensile strength	●○○○	●●●○	●●●●
Light weight	●●●●	●●●●	●●○○
Corrosion resistance	●●●○	●●●●	●●○○
Typical use	Short spans, urban networks, busbars	Distribution & transmission, coastal / desert	Long spans, transmission, crossings

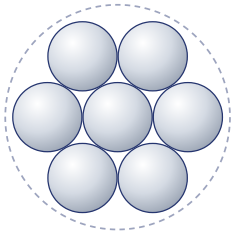
Relative comparison for equal conductor diameter — for guidance only.

TYPE AAC

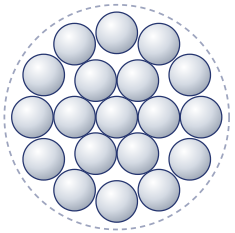
All Aluminium Conductor

موصلات الألومنيوم

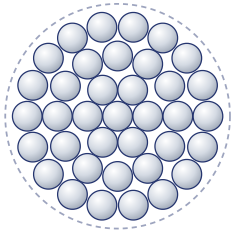
AAC conductors are made of hard-drawn electrical-grade aluminium wires (AL1 / 1350, min. 99.5 % Al) concentrically stranded in one or more layers. They offer the highest conductivity-to-weight ratio of any bare conductor and inherently good corrosion resistance.



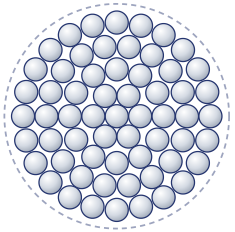
7 wires



19 wires



37 wires



61 wires

Applications

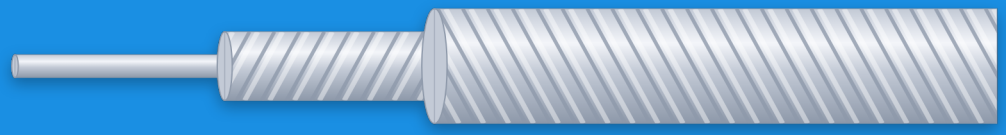
- Urban and rural MV/LV distribution lines
- Short spans where high current is needed
- Substation busbars and jumpers
- Coastal areas with moderate corrosion

Construction

- Aluminium wires AL1 to IEC 60889
- 7, 19, 37, 61 or 91 wires
- Alternate-direction layers, right-hand outer layer
- Optional grease protection to EN 50326

Standards

IEC 61089 | BS EN 50182 | IEC 60889 | ES



CONSTRUCTION VIEW — LAYERS STEPPED BACK TO SHOW ALTERNATE LAY DIRECTIONS

AAC · METRIC SIZES

All Aluminium Conductor — Metric Series

to IEC 61089 / BS EN 50182 (DIN series designations) and ES

موصلات الألومنيوم

Product code	Nominal area	Calculated area	Stranding	Overall diameter	Approx. mass	DC resistance at 20 °C	Rated tensile strength	Current rating
	mm²	mm²	No. × mm	mm	kg/km	Ω/km	kN	A
ATC-AAC-001	**16**	15.9	7 × 1.70	5.1	43.4	1.7986	3.02	**110**
ATC-AAC-002	**25**	24.2	7 × 2.10	6.3	66.3	1.1787	4.36	**140**
ATC-AAC-003	**35**	34.4	7 × 2.50	7.5	93.9	0.8317	6.01	**175**
ATC-AAC-004	**50**	49.5	7 × 3.00	9.0	135.2	0.5776	8.41	**225**
ATC-AAC-005	**50**	48.3	19 × 1.80	9.0	132.9	0.5944	8.94	**220**
ATC-AAC-006	**70**	65.8	19 × 2.10	10.5	180.9	0.4367	11.85	**265**
ATC-AAC-007	**95**	93.3	19 × 2.50	12.5	256.3	0.3081	16.32	**330**
ATC-AAC-008	**120**	117.0	19 × 2.80	14.0	321.5	0.2456	19.89	**385**
ATC-AAC-009	**150**	147.1	37 × 2.25	15.8	405.7	0.1960	26.48	**445**
ATC-AAC-010	**185**	181.6	37 × 2.50	17.5	500.9	0.1588	31.78	**505**
ATC-AAC-011	**240**	242.5	61 × 2.25	20.3	671.1	0.1193	43.66	**605**
ATC-AAC-012	**300**	299.4	61 × 2.50	22.5	828.5	0.0966	52.40	**690**
ATC-AAC-013	**400**	400.1	61 × 2.89	26.0	1107.1	0.0723	68.02	**830**
ATC-AAC-014	**500**	499.8	61 × 3.23	29.1	1382.9	0.0579	82.47	**960**
ATC-AAC-015	**625**	626.2	91 × 2.96	32.6	1739.7	0.0464	106.45	**1105**
ATC-AAC-016	**800**	802.1	91 × 3.35	36.9	2228.3	0.0362	132.34	**1290**
ATC-AAC-017	**1000**	999.7	91 × 3.74	41.1	2777.3	0.0291	159.95	**1485**

Values are nominal unless stated. Mass is approximate. Current ratings are calculated per IEEE 738 for 50 Hz, ambient 35 °C, conductor 80 °C, wind 0.6 m/s, solar radiation 900 W/m², emissivity and absorptivity 0.5, at sea level — for guidance only. Other sizes, standards and product codes on request.

All Aluminium Conductor — BS EN 50182 (UK)

code-name sizes to BS EN 50182

موصلات الألومنيوم

Product code	Code name	Designation	Area	Stranding	Overall diameter	Approx. mass	DC resistance at 20 °C	Rated tensile strength	Current rating
		BS EN 50182	mm²	No. × mm	mm	kg/km	Ω/km	kN	A
ATC-AAC-B01	Midge	23-AL1	23.30	7 × 2.06	6.18	63.8	1.225	4.20	140
ATC-AAC-B02	Gnat	27-AL1	26.90	7 × 2.21	6.63	73.4	1.064	4.83	150
ATC-AAC-B03	Mosquito	37-AL1	36.90	7 × 2.59	7.77	100.8	0.775	6.27	185
ATC-AAC-B04	Ladybird	43-AL1	42.80	7 × 2.79	8.37	117.0	0.668	7.28	205
ATC-AAC-B05	Ant	53-AL1	52.80	7 × 3.10	9.30	144.4	0.541	8.72	230
ATC-AAC-B06	Fly	64-AL1	63.60	7 × 3.40	10.20	173.7	0.4497	10.49	260
ATC-AAC-B07	Bluebottle	74-AL1	73.60	7 × 3.66	11.00	201.3	0.388	11.78	285
ATC-AAC-B08	Earwig	79-AL1	78.60	7 × 3.78	11.30	214.7	0.364	12.57	295
ATC-AAC-B09	Grasshopper	84-AL1	84.10	7 × 3.91	11.70	229.7	0.34	13.45	310
ATC-AAC-B10	Clegg	96-AL1	95.60	7 × 4.17	12.50	261.3	0.299	15.30	335
ATC-AAC-B11	Wasp	106-AL1	106.00	7 × 4.39	13.20	289.6	0.2697	16.95	360
ATC-AAC-B12	Beetle	106-AL1	106.40	19 × 2.67	13.40	292.4	0.2701	18.08	360
ATC-AAC-B13	Bee	132-AL1	132.00	7 × 4.90	14.70	360.8	0.217	21.12	415
ATC-AAC-B14	Hornet	158-AL1	157.60	19 × 3.25	16.30	433.2	0.182	26.01	465
ATC-AAC-B15	Caterpillar	186-AL1	185.90	19 × 3.53	17.70	511.1	0.155	29.75	515
ATC-AAC-B16	Chafer	213-AL1	213.20	19 × 3.78	18.90	586	0.135	34.12	560
ATC-AAC-B17	Spider	238-AL1	237.60	19 × 3.99	20.00	652.9	0.121	38.01	600
ATC-AAC-B18	Cockroach	266-AL1	265.70	19 × 4.22	21.10	730.4	0.108	42.52	645
ATC-AAC-B19	Butterfly	323-AL1	322.70	19 × 4.65	23.30	886.8	0.089	51.63	730
ATC-AAC-B20	Moth	373-AL1	373.10	19 × 5.00	25.00	1025.3	0.077	59.69	800
ATC-AAC-B21	Drone	372-AL1	372.40	37 × 3.58	25.10	1027.1	0.0774	59.59	795
ATC-AAC-B22	Centipede	415-AL1	415.20	37 × 3.78	26.50	1145.1	0.0695	66.43	855
ATC-AAC-B23	Maybug	486-AL1	486.10	37 × 4.09	28.60	1340.6	0.0593	77.78	940
ATC-AAC-B24	Scorpion	530-AL1	529.80	37 × 4.27	29.90	1461.2	0.0544	84.77	995
ATC-AAC-B25	Cicada	628-AL1	628.30	37 × 4.65	32.60	1732.9	0.0459	100.54	1110

Values are nominal unless stated. Mass is approximate. Current ratings are calculated per IEEE 738 for 50 Hz, ambient 35 °C, conductor 80 °C, wind 0.6 m/s, solar radiation 900 W/m², emissivity and absorptivity 0.5, at sea level — for guidance only. Other sizes, standards and product codes on request.

TYPE AAAC

All Aluminium Alloy Conductor

موصلات سبائك الألومنيوم

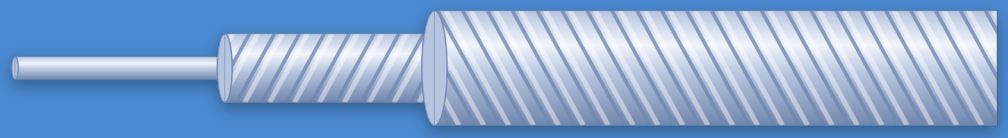
AAAC conductors are stranded from heat-treated aluminium-magnesium-silicon alloy wires (AL3 / 6201-T81). They combine high mechanical strength with good conductivity, giving lower sag and longer spans than AAC — without the corrosion risk of a steel core.

Applications

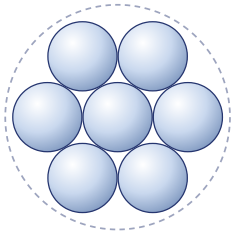
- MV distribution and HV transmission lines
- Long spans and reduced tower count
- Coastal, desert and industrial environments
- Re-conductoring of existing ACSR routes

Construction

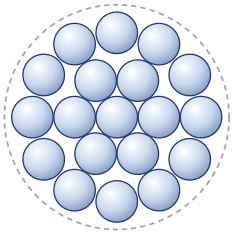
- Al-Mg-Si alloy wires AL3 to EN 50183 / IEC 60104
- 7, 19, 37, 61 or 91 wires
- Alternate-direction layers, right-hand outer layer
- Higher strength-to-weight ratio than ACSR



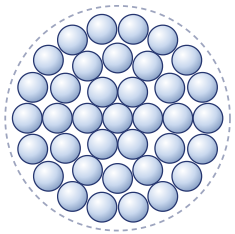
CONSTRUCTION VIEW — LAYERS STEPPED BACK TO SHOW ALTERNATE LAY DIRECTIONS



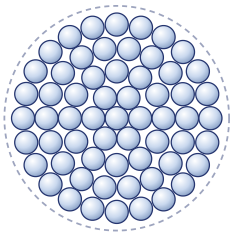
7 wires



19 wires



37 wires



61 wires

Standards

IEC 61089 | BS EN 50182 | EN 50183 | ES

AAAC · METRIC SIZES

All Aluminium Alloy Conductor — Metric Series

to IEC 61089 / BS EN 50182 (DIN series designations) and ES

موصلات سبائك الألومنيوم

Product code	Nominal area	Calculated area	Stranding	Overall diameter	Approx. mass	DC resistance at 20 °C	Rated tensile strength	Current rating
	mm²	mm²	No. × mm	mm	kg/km	Ω/km	kN	A
ATC-AAA-001	**16**	15.9	7 × 1.70	5.1	43.4	2.0701	4.69	**105**
ATC-AAA-002	**25**	24.2	7 × 2.10	6.3	66.2	1.3566	7.15	**135**
ATC-AAA-003	**35**	34.4	7 × 2.50	7.5	93.8	0.9572	10.14	**165**
ATC-AAA-004	**50**	49.5	7 × 3.00	9.0	135.1	0.6647	14.60	**210**
ATC-AAA-005	**50**	48.3	19 × 1.80	9.0	132.7	0.6841	14.26	**205**
ATC-AAA-006	**70**	65.8	19 × 2.10	10.5	180.7	0.5026	19.41	**250**
ATC-AAA-007	**95**	93.3	19 × 2.50	12.5	256.0	0.3546	27.51	**315**
ATC-AAA-008	**120**	117.0	19 × 2.80	14.0	321.2	0.2827	34.51	**360**
ATC-AAA-009	**150**	147.1	37 × 2.25	15.8	405.3	0.2256	43.40	**415**
ATC-AAA-010	**185**	181.6	37 × 2.50	17.5	500.3	0.1827	53.58	**475**
ATC-AAA-011	**240**	242.5	61 × 2.25	20.3	670.3	0.1373	71.55	**570**
ATC-AAA-012	**300**	299.4	61 × 2.50	22.5	827.5	0.1112	88.33	**650**
ATC-AAA-013	**400**	400.1	61 × 2.89	26.0	1105.9	0.0832	118.04	**785**
ATC-AAA-014	**500**	499.8	61 × 3.23	29.1	1381.4	0.0666	147.45	**905**
ATC-AAA-015	**625**	626.2	91 × 2.96	32.6	1737.7	0.0534	184.73	**1040**
ATC-AAA-016	**800**	802.1	91 × 3.35	36.9	2225.8	0.0417	236.62	**1215**
ATC-AAA-017	**1000**	999.7	91 × 3.74	41.1	2774.3	0.0334	294.91	**1400**

Values are nominal unless stated. Mass is approximate. Current ratings are calculated per IEEE 738 for 50 Hz, ambient 35 °C, conductor 80 °C, wind 0.6 m/s, solar radiation 900 W/m², emissivity and absorptivity 0.5, at sea level — for guidance only. Other sizes, standards and product codes on request.

AAAC · BS CODE NAMES

All Aluminium Alloy Conductor — BS EN 50182 (UK)

code-name sizes to BS EN 50182

موصلات سبائك الألومنيوم

Product code	Code name	Designation	Area	Stranding	Overall diameter	Approx. mass	DC resistance at 20 °C	Rated tensile strength	Current rating
		BS EN 50182	mm²	No. × mm	mm	kg/km	Ω/km	kN	A
ATC-AAA-B01	Box	19-AL3	18.80	7 × 1.85	5.55	51.40	1.748	5.55	115
ATC-AAA-B02	Acacia	24-AL3	23.80	7 × 2.08	6.24	64.90	1.3828	7.02	130
ATC-AAA-B03	Almond	30-AL3	30.10	7 × 2.34	7.02	82.20	1.0926	8.88	155
ATC-AAA-B04	Cedar	35-AL3	35.50	7 × 2.54	7.62	96.80	0.9273	10.46	170
ATC-AAA-B05	Deodar	42-AL3	42.20	7 × 2.77	8.31	115.20	0.7797	12.44	190
ATC-AAA-B06	Fir	48-AL3	47.80	7 × 2.95	8.85	130.60	0.6875	14.11	205
ATC-AAA-B07	Hazel	60-AL3	59.90	7 × 3.30	9.90	163.40	0.5494	17.66	235
ATC-AAA-B08	Pine	72-AL3	71.60	7 × 3.61	10.80	195.60	0.4591	21.14	265
ATC-AAA-B09	Holly	84-AL3	84.10	7 × 3.91	11.70	229.50	0.3913	24.79	295
ATC-AAA-B10	Willow	90-AL3	89.70	7 × 4.04	12.10	245.00	0.3665	26.47	305
ATC-AAA-B11	Oak	119-AL3	118.90	7 × 4.65	14.00	324.50	0.2767	35.07	365
ATC-AAA-B12	Mulberry	151-AL3	150.90	19 × 3.18	15.90	414.30	0.2192	44.52	425
ATC-AAA-B13	Ash	181-AL3	180.70	19 × 3.48	17.40	496.10	0.183	53.31	475
ATC-AAA-B14	Elm	211-AL3	211.00	19 × 3.76	18.80	579.20	0.1568	62.24	525
ATC-AAA-B15	Poplar	239-AL3	239.40	37 × 2.87	20.10	659.40	0.1387	70.61	565
ATC-AAA-B16	Sycamore	303-AL3	303.20	37 × 3.23	22.60	835.20	0.1095	89.40	660
ATC-AAA-B17	Upas	362-AL3	362.10	37 × 3.53	24.70	997.50	0.0917	106.82	735
ATC-AAA-B18	Yew	479-AL3	479.00	37 × 4.06	28.40	1319.60	0.0693	141.31	880
ATC-AAA-B19	Totara	498-AL3	498.10	37 × 4.14	29.00	1372.10	0.0666	146.93	900
ATC-AAA-B20	Rubus	587-AL3	586.90	61 × 3.50	31.50	1622.00	0.0567	173.13	1000
ATC-AAA-B21	Sorbus	659-AL3	659.40	61 × 3.71	33.40	1822.50	0.0508	194.53	1075
ATC-AAA-B22	Araucaria	821-AL3	821.10	61 × 4.14	37.30	2269.40	0.0406	242.24	1235
ATC-AAA-B23	Redwood	996-AL3	996.20	61 × 4.56	41.00	2753.20	0.0334	293.88	1400

Values are nominal unless stated. Mass is approximate. Current ratings are calculated per IEEE 738 for 50 Hz, ambient 35 °C, conductor 80 °C, wind 0.6 m/s, solar radiation 900 W/m², emissivity and absorptivity 0.5, at sea level — for guidance only. Other sizes, standards and product codes on request.

TYPE ACSR

Aluminium Conductor Steel Reinforced

موصلات الألومنيوم المقواة بالصلب

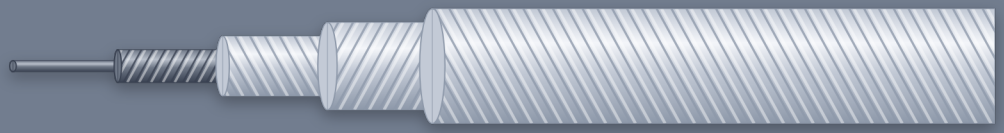
ACSR conductors consist of one or more layers of hard-drawn aluminium wires (AL1) stranded over a core of galvanized high-strength steel wires (ST1A). The steel core carries the mechanical load while the aluminium carries the current — the proven choice for long spans and heavy loading.

Applications

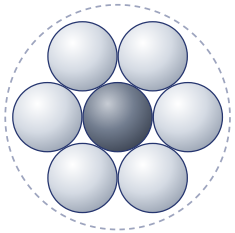
- HV and EHV transmission lines
- MV distribution lines with long spans
- River and valley crossings
- Areas with high wind or ice loading

Construction

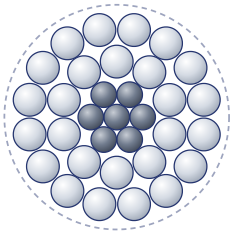
- Aluminium wires AL1 to IEC 60889
- Galvanized steel core ST1A, class A zinc coating
- Steel core of 1 or 7 wires
- Core or full-body grease to EN 50326 on request



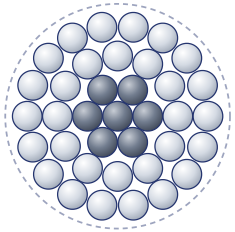
CONSTRUCTION VIEW — LAYERS STEPPED BACK TO SHOW ALTERNATE LAY DIRECTIONS



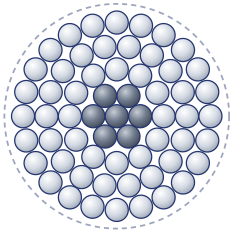
6/1



26/7



30/7



54/7

Standards

IEC 61089 | BS EN 50182 | EN 50189 | ES

Aluminium Conductor Steel Reinforced — Metric Series

to IEC 61089 / BS EN 50182 (DIN series designations) and ES

موصلات الألومنيوم المقواة بالصلب

Product code	Designation	Area Al	Area steel	Total area	Stranding Al	Stranding steel	Overall diameter	Approx. mass	DC resistance at 20 °C	Rated tensile strength	Current rating
	Al / St	mm²	mm²	mm²	No. × mm	No. × mm	mm	kg/km	Ω/km	kN	A
ATC-ACS-001	16/2.5	15.3	2.5	17.8	6×1.80	1×1.80	5.4	62	1.8769	5.80	110
ATC-ACS-002	25/4	23.9	4.0	27.8	6×2.25	1×2.25	6.8	96	1.2012	8.95	145
ATC-ACS-003	35/6	34.4	5.7	40.1	6×2.70	1×2.70	8.1	139	0.8342	12.37	180
ATC-ACS-004	50/8	48.3	8.0	56.3	6×3.20	1×3.20	9.6	195	0.5939	16.81	225
ATC-ACS-005	70/12	69.9	11.4	81.3	26×1.85	7×1.44	11.7	282	0.4132	26.27	280
ATC-ACS-006	95/15	94.4	15.3	109.7	26×2.15	7×1.67	13.6	380	0.3060	34.93	340
ATC-ACS-007	120/20	121.6	19.8	141.4	26×2.44	7×1.90	15.5	490	0.2376	44.50	400
ATC-ACS-008	150/25	148.9	24.2	173.1	26×2.70	7×2.10	17.1	600	0.1940	53.67	455
ATC-ACS-009	185/30	183.8	29.8	213.6	26×3.00	7×2.33	19.0	740	0.1571	65.27	520
ATC-ACS-010	210/35	209.1	34.1	243.2	26×3.20	7×2.49	20.3	843	0.1381	73.36	565
ATC-ACS-011	240/40	243.1	39.5	282.5	26×3.45	7×2.68	21.8	979	0.1188	85.12	620
ATC-ACS-012	380/50	381.7	49.5	431.2	54×3.00	7×3.00	27.0	1443	0.0758	121.30	820
ATC-ACS-013	490/65	490.3	63.6	553.8	54×3.40	7×3.40	30.6	1851	0.0590	150.81	960

Values are nominal unless stated. Mass is approximate. Current ratings are calculated per IEEE 738 for 50 Hz, ambient 35 °C, conductor 80 °C, wind 0.6 m/s, solar radiation 900 W/m², emissivity and absorptivity 0.5, at sea level — for guidance only. Other sizes, standards and product codes on request.

Aluminium Conductor Steel Reinforced — BS EN 50182 (UK)

code-name sizes to BS EN 50182 · Mole to Coyote

موصلات الألومنيوم المقواة بالصلب

Product code	Code name	Area Al	Area steel	Total area	Stranding Al	Stranding steel	Overall diameter	Approx. mass	DC resistance at 20 °C	Rated tensile strength	Current rating
		mm²	mm²	mm²	No. × mm	No. × mm	mm	kg/km	Ω/km	kN	A
ATC-ACS-B01	Mole	10.60	1.77	12.4	6×1.50	1×1.50	4.50	42.8	2.7027	4.14	85
ATC-ACS-B02	Squirrel	21.00	3.50	24.5	6×2.11	1×2.11	6.33	84.7	1.3659	7.87	130
ATC-ACS-B03	Gopher	26.20	4.37	30.6	6×2.36	1×2.36	7.08	105.9	1.0919	9.58	150
ATC-ACS-B04	Weasel	31.60	5.27	36.9	6×2.59	1×2.59	7.77	127.6	0.9065	11.38	170
ATC-ACS-B05	Fox	36.70	6.11	42.8	6×2.79	1×2.79	8.37	148.1	0.7812	13.21	190
ATC-ACS-B06	Ferret	42.40	7.07	49.5	6×3.00	1×3.00	9.00	171.2	0.6757	15.27	205
ATC-ACS-B07	Rabbit	52.90	8.81	61.7	6×3.35	1×3.35	10.05	213.5	0.5419	18.42	235
ATC-ACS-B08	Mink	63.10	10.50	73.6	6×3.66	1×3.66	11.00	254.8	0.4540	21.67	265
ATC-ACS-B09	Skunk	63.20	36.90	100.1	12×2.59	7×2.59	12.95	463.0	0.4568	52.79	275
ATC-ACS-B10	Beaver	75.00	12.50	87.5	6×3.99	1×3.99	12.00	302.9	0.3820	25.76	295
ATC-ACS-B11	Horse	73.40	42.80	116.2	12×2.79	7×2.79	13.95	537.3	0.3936	61.26	300
ATC-ACS-B12	Racoon	78.80	13.10	91.9	6×4.09	1×4.09	12.30	318.3	0.3635	27.06	305
ATC-ACS-B13	Otter	83.90	14.00	97.9	6×4.22	1×4.22	12.70	338.8	0.3415	28.81	315
ATC-ACS-B14	Cat	95.40	15.90	111.3	6×4.50	1×4.50	13.50	385.3	0.3003	32.76	345
ATC-ACS-B15	Hare	105.00	17.50	122.5	6×4.72	1×4.72	14.20	423.8	0.2730	36.04	365
ATC-ACS-B16	Dog	105.00	13.60	118.6	6×4.72	7×1.57	14.15	394.0	0.2733	32.65	365
ATC-ACS-B17	Coyote	131.70	20.10	151.8	26×2.54	7×1.91	15.90	520.7	0.2192	45.86	420
ATC-ACS-B18	Cougar	131.50	7.31	138.8	18×3.05	1×3.05	15.25	418.8	0.2188	29.74	415

Values are nominal unless stated. Mass is approximate. Current ratings are calculated per IEEE 738 for 50 Hz, ambient 35 °C, conductor 80 °C, wind 0.6 m/s, solar radiation 900 W/m², emissivity and absorptivity 0.5, at sea level — for guidance only. Other sizes, standards and product codes on request.

Aluminium Conductor Steel Reinforced — BS EN 50182 (UK)

code-name sizes to BS EN 50182 · Cougar to Moose

موصلات الألومنيوم المقواة بالصلب

Product code	Code name	Area Al	Area steel	Total area	Stranding Al	Stranding steel	Overall diameter	Approx. mass	DC resistance at 20 °C	Rated tensile strength	Current rating
		mm²	mm²	mm²	No. × mm	No. × mm	mm	kg/km	Ω/km	kN	A
ATC-ACS-B19	Tiger	131.20	30.60	161.8	30×2.36	7×2.36	16.50	602.2	0.2202	57.87	420
ATC-ACS-B20	Wolf	158.10	36.90	195.0	30×2.59	7×2.59	18.10	725.3	0.1829	68.91	475
ATC-ACS-B21	Dingo	158.70	8.81	167.5	18×3.35	1×3.35	16.75	505.2	0.1814	35.87	465
ATC-ACS-B22	Lynx	183.40	42.80	226.2	30×2.79	7×2.79	19.50	841.6	0.1576	79.97	520
ATC-ACS-B23	Caracal	184.20	10.20	194.4	18×3.61	1×3.61	18.05	586.7	0.1562	40.74	515
ATC-ACS-B24	Panther	212.10	49.50	261.6	30×3.00	7×3.00	21.00	973.1	0.1363	92.46	570
ATC-ACS-B25	Jaguar	210.60	11.70	222.3	18×3.86	1×3.86	19.30	670.8	0.1366	46.57	560
ATC-ACS-B26	Lion	238.30	55.60	293.9	30×3.18	7×3.18	22.30	1093.4	0.1213	100.47	615
ATC-ACS-B27	Bear	264.40	61.70	326.1	30×3.35	7×3.35	23.50	1213.4	0.1093	111.50	660
ATC-ACS-B28	Goat	324.30	75.70	400.0	30×3.71	7×3.71	26.00	1488.2	0.0891	135.13	750
ATC-ACS-B29	Sheep	375.10	87.50	462.6	30×3.99	7×3.99	27.90	1721.3	0.0771	156.30	820
ATC-ACS-B30	Antelope	374.10	48.50	422.6	54×2.97	7×2.97	26.70	1413.8	0.0773	118.88	810
ATC-ACS-B31	Bison	381.70	49.50	431.2	54×3.00	7×3.00	27.00	1442.5	0.0758	121.30	820
ATC-ACS-B32	Deer	429.60	100.20	529.8	30×4.27	7×4.27	29.90	1971.4	0.0673	179.00	895
ATC-ACS-B33	Zebra	428.90	55.60	484.5	54×3.18	7×3.18	28.60	1620.8	0.0674	131.92	885
ATC-ACS-B34	Elk	477.10	111.30	588.4	30×4.50	7×4.50	31.50	2189.4	0.0606	198.80	955
ATC-ACS-B35	Camel	476.00	61.70	537.7	54×3.35	7×3.35	30.20	1798.8	0.0608	146.40	945
ATC-ACS-B36	Moose	528.50	68.50	597.0	54×3.53	7×3.53	31.80	1997.3	0.0547	159.92	1010

Values are nominal unless stated. Mass is approximate. Current ratings are calculated per IEEE 738 for 50 Hz, ambient 35 °C, conductor 80 °C, wind 0.6 m/s, solar radiation 900 W/m², emissivity and absorptivity 0.5, at sea level — for guidance only. Other sizes, standards and product codes on request.

REFERENCE

Technical Data

بيانات فنية

Wire material properties

Property	Unit	Aluminium AL1	Alloy AL3	Steel ST1A
Standard		IEC 60889	EN 50183	EN 50189
Resistivity at 20 °C	nΩ·m	28.264	32.53	191.57
Conductivity at 20 °C	% IACS	61.0	53.0	9.0
Density	kg/dm³	2.703	2.703	7.78
Temp. coefficient of resistance	1/K	0.00403	0.0036	0.0044
Coefficient of linear expansion	1/K	23×10^{-6}	23×10^{-6}	11.5×10^{-6}

Resistance at operating temperature

$$R_T = R_{20} \times [\, 1 + \alpha_{20} \times (T - 20) \,]$$

R_{20} — DC resistance at 20 °C from the data tables (Ω/km); α_{20} — temperature coefficient (0.00403 /K for AAC and ACSR, 0.0036 /K for AAAC); T — conductor temperature (°C). AC resistance at 50 Hz is slightly higher due to skin effect.

Unit conversions

1 mm²	= 1.9735 kcmil
1 mm	= 0.03937 in
1 kg/km	= 0.6720 lb/kft
1 Ω/km	= 0.3048 Ω/kft
1 kN	= 101.97 kgf · 224.81 lbf
1 km	= 3280.84 ft

Current rating basis

IEEE 738 steady-state heat balance · 50 Hz · ambient 35 °C · conductor 80 °C · wind 0.6 m/s across the conductor · solar 900 W/m² · ε = α = 0.5 · sea level.

DELIVERY

Packing & Drums

التعبئة والبكرات

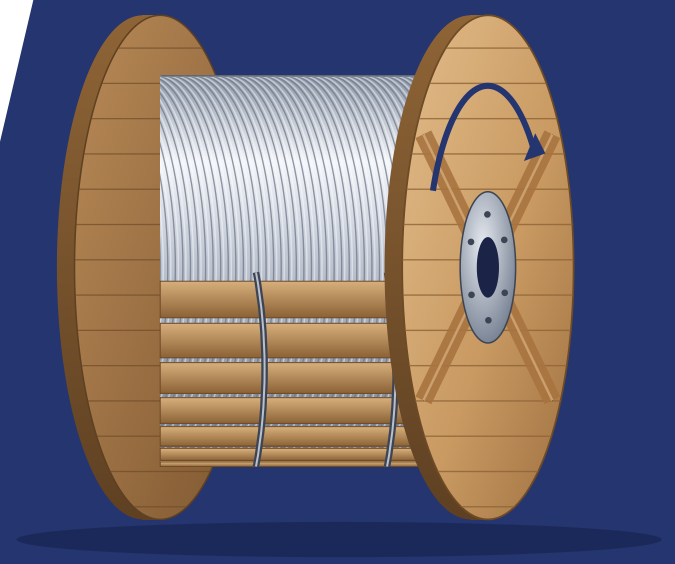
Conductors are wound under controlled tension onto drums sized for the conductor diameter and the ordered length, protecting the conductor from damage during transport, storage and stringing.

Drum options

- Non-returnable wooden drums, treated for export
- Returnable steel drums for large sizes and long lengths
- Standard or customer-specified delivery lengths
- Inner and outer conductor ends firmly secured
- Outer layer protected by wooden lagging or wrapping

Drum marking

- Manufacturer's name and drum number
- Conductor type, size / code name and standard
- Length (m), net and gross mass (kg)
- Production lot and date · direction-of-rolling arrow



D1 flange diameter · **D2** barrel diameter · **d** arbor hole
B traverse (inside width) · **A** overall width
Drum size is selected to suit the conductor diameter and length — drum schedule on request.

HOW TO ORDER

Ordering Information

معلومات الطلب

To receive an accurate quotation, please include the following in your enquiry:

01	**Conductor type**	AAC, AAAC or ACSR
02	**Size**	Nominal area, designation or code name (e.g. 120/20, Lynx)
03	**Standard**	IEC 61089, BS EN 50182, ES or customer specification
04	**Quantity**	Total length (km) and length per drum
05	**Drum type**	Wooden or steel
06	**Grease (if any)**	Case per EN 50326 — ACSR and AAC
07	**Inspection**	Witness tests or third-party inspection, if required

Example: ACSR 240/40 to BS EN 50182 · 60 km in 2,000 m lengths · non-returnable wooden drums · core greased.

Product code structure

ATC-ACS-011

ATC	Altawakol for Cables
AAC / AAA / ACS	Conductor family
001 …	Metric series item
B01 …	BS EN 50182 code-name item

Quote the product code from the data tables to identify the exact construction.

Thank you

تواصل معنا

ALTAWAKOL CABLES

ALTAWAKOL FOR CABLES

إحدى شركات مجموعة التوكل

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